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Answer following question
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1:- What is 2 tailed and tailed T Test ?

Ans :- The t-test is used to test hypotheses about a population mean.

1. One-Tailed t-Test

A one-tailed test checks for a difference in only one direction (either greater than or less than).

Types:

Right-tailed test:

H0 :μ = μ0$H$1: μ > μ0 H1:μ>μ0

Left-tailed test:

H0: μ = μ0 H0 :μ=μ 0$H$1: μ < μ 0H 1 :μ < μ02. Two-Tailed t-Test A two-tailed test checks for a difference in both directions (either greater or less). Hypothesis H 0

:μ=μ 0

H 1 : μ ≠ μ 0 H 1

:μ □ =μ 0

2 :- Explain what is Type I and Type II errors with an example.

Ans :- Type I and Type II Errors

In hypothesis testing, errors may occur while making decisions about the null hypothesis (H_0). These errors are classified as Type I error and Type II error.

1. Type I Error (α Error)

Definition: Type I error occurs when the null hypothesis (H_0) is rejected even though it is true.

Symbol: α (level of significance)

Explanation:

It means we conclude that there is a significant effect when actually there is none.

Example:

Let, H_0: Average age of students = 19

If the true average age is 19, but we reject H_0, then it is a Type I error.

Real-life example: A medical test shows a person is diseased, but actua

2. Type II Error (β Error)

Definition: Type II error occurs when the null hypothesis (H_0) is not rejected even though it is false.

Symbol: β

Explanation: It means we fail to detect a real effect.

Example: Let, H_0: Average age of students = 19

If the actual average age is not 19, but we accept H_0, then it is a Type II error.

Real-life example: A medical test shows a person is healthy, but actually, the person has a disease.

3 :- What is power of statistical test?

Ans :- Power of Statistical Test

Definition: The power of a statistical test is the probability of rejecting the null hypothesis (H_0) when it is actually false.

Formula: Power = 1 – β (where β = probability of Type II error)

Explanation:

Power measures the ability of a test to correctly detect a real effect. A higher power means the test is more effective in identifying false null hypotheses.

Example:

Let, H_0: Average marks of students = 60

If the actual average marks are not 60, and the test correctly rejects H_0, then the test has high power.

Conclusion:

A good statistical test should have high power, meaning it should minimize Type II error (β).

4 :- Explain input and out put characteristics for stats.ttest_rel function in your own words

Ans :- Input and Output Characteristics of stats.ttest_rel()

The stats.ttest_rel() function (from the SciPy library) is used to perform a paired t-test. It compares two related samples to check whether their means are significantly different.

Input Characteristics

1 :- Two Related Samples (a, b):

The function takes two arrays/lists of data.

Both samples must be dependent (paired).

Example: marks of the same students before and after an exam.

2:- Same Size Requirement:

Both input samples must have equal number of observations.

3 :- Numeric Data:

Inputs should contain numerical values (integers or floats).

4 :- Optional Parameters:

axis: defines how the test is applied across data.

nan_policy: handles missing values (ignore, raise error, etc.).

alternative: specifies hypothesis type (two-sided, greater, less).

Output Characteristics

The function returns two main values:

1 :- t-statistic:

Measures the difference between paired samples.

Larger value ⇒ greater difference.

2 :- p-value:

Helps in decision making.

If p-value < significance level (α) ⟶ Reject H_0

If p-value ≥ α ⟶ Accept H_0

Conclusion

stats.ttest_rel() takes paired numerical data as input and gives t-statistic and p-value as output to determine whether there is a significant difference between the two related samples.

5 :- What is function in stats module for identifying power of a statistical test. Explain out put range of it and its significance.

Ans :- Function Description

ttest_power() is used to compute the power of a t-test, i.e., the probability of correctly rejecting the null hypothesis (H_0) when it is false.

Output Range The output of ttest_power() lies between 0 and 1.

Interpretation:

Power = 0 ⟶ Test never detects a true effect (very poor test) Power = 1 ⟶ Test always detects a true effect (ideal test) Power ≥ 0.8 ⟶ Considered a good and reliable test

Significance of Power Measures Test Effectiveness: Higher power means the test is better at detecting real differences.

Reduces Type II Error (β): Power = 1 − β, so higher power means lower chance of missing a true effect.

Helps in Sample Size Planning: Researchers use power to decide the required sample size for accurate results.

Improves Decision Making: Ensures that important effects are not ignored.

Conclusion

The ttest_power() function helps evaluate how strong a statistical test is. A higher power indicates a more reliable test with a lower probability of Type II error.